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Transfer contrastive learning for Raman spectra data of urine: Detection of glucose, protein, and prediction of kidney disorders

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ABSTRACT

Raman spectroscopy, a non-invasive analytical technique, reveals significant potential in clinical diagnosis of kidney disorders by detecting key biomolecules in urine samples, especially glucose and protein. Although machine learning models have been widely applied for efficiently analyzing Raman spectral data, the high-dimensionality, imbalance and sample-scarcity of Raman spectral data still pose challenges to the models in achieving accurate detection. To address these challenges, we propose a novel deep learning model, TCRaman, which integrates transfer learning and contrastive learning for urine detection using Raman spectral data. As contrastive learning is capable of representation learning on imbalanced data, TCRaman first utilizes a pretrained contrastive learning model on a large labeled Raman spectral dataset of bacteria, to enhance the model's capability to learn meaningful low-dimensional representations from high-dimensional Raman spectral data. Then, the pretrained model is finetuned on clinical urine Raman spectral data. This transfer learning framework is a foundation model that can break through the limitation of sample-scarcity on different downstream tasks. The experiments demonstrate the superiority of TCRaman compared with current state-of-the-art models. The results show that TCRaman achieves 91% accuracy on the detection of both glucose and protein, and 95% accuracy on the prediction of kidney disorders, highlighting the effectiveness of our proposed method in detecting urine Raman spectra. The proposed TCRaman method provides a promising way for accurate, rapid, and cost-effective detection for spectral data of biochemical samples.

1. Introduction

The detection of glucose and protein in urine is an important way for clinical diagnosis and monitoring of many diseases, especially kidney disorders and diabetes [1–3]. Glucose detection in urine is a hallmark of hyperglycemia, often associated with diabetes and diabetes-related complications, including retinopathy, kidney disorders, cardiovascular disease and so on [4,5]. On the other hand, protein is normally restricted within the bloodstream due to the selective permeability of the glomerular membrane, thus indicative of kidney injury when detected in urine, warranting further medical evaluation [6,7]. Thus, the glucose and protein detection in urine lie in the ability to serve as early warning signals for underlying health issues, especially kidney disorders. Therefore, various techniques are applied for detecting glucose and protein in urine to assist the diagnosis, such as chromatography [8], mass spectrometry [9], infrared spectroscopy [10], and Raman spectroscopy [11]. Raman spectroscopy provides a non-invasive, fast, and low-cost detection [12–15], making it a valuable technique for clinical diagnosis and disease monitoring [16–18]. Previous studies have applied urine Raman spectroscopy to the diagnosis of kidney disorders [19,20], diabetes [21], bladder cancer [22] and so on. However, these methods can only focus on the information of some specific peaks in the Raman spectrum, but ignore the hidden patterns in the complex and high-dimensional Raman spectral data. Therefore, machine learning models have been widely applied to overcome the complexity and high-dimensionality of Raman spectra.

Currently, machine learning combined with dimension reduction algorithms, such as principal component analysis or linear discriminant analysis, have achieved high discrimination accuracy for Raman

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spectra of biochemical samples [23,24], and are also applied for detection of urinary Raman spectra to detect key biomolecules in urine, such as protein, urea, creatinine, and glucose [25,26]. The detection of these biomolecules in urine can be adopted for kidney disorders predicting by machine learning models, such as orthogonal partial least squares discriminant analysis [27], support vector machine [28] and partial least squares regression [29]. Compared with traditional machine learning models, deep learning reveals the superiority for dimension reduction of Raman spectral data [30–33]. Hence, the end-to-end deep learning models have been widely applied for biochemical Raman spectroscopy, for instance, bacterial Raman spectra identification, including convolutional neural networks (CNN) [34], long-short term memory (LSTM) [35], scale-adaptive neural networks [36] and so on. Recently, convolutional neural networks [37,38] are also adopted for kidney disorders diagnosis using Raman spectral data of urine.

However, the application of machine learning (especially deep learning) models to biochemical Raman spectroscopy detection faces the challenge from the scarcity of labeled data. Therefore, the transfer learning of deep learning models emerges as a promising solution. Transfer learning first pretrain neural networks on large-scale well-annotated datasets, and then finetune the pretrained model in downstream tasks [39–41]. Notably, due to the universality of data scarcity across different tasks in chemistry, it is important to transfer models pretrained on large datasets to multiple downstream tasks for fine-tuning. This framework has been applied to chemical molecules [42] and hyperspectral image data [43]. However, current studies about transfer learning on Raman spectral data [44–46] only adopt pretrained model for a single downstream task, since transfer learning is still insufficient to learn low-dimensional representations from complex and high-dimensional Raman spectral data under multi-tasks. On the other hand, imbalanced data limit the performance of transfer learning model on Raman spectral data, which are widely presented in biochemical Raman spectroscopy. For instance, on the detection of glucose and protein in urine Raman spectroscopy, negative (healthy) samples are much more than positive (patient) samples.

To address this issue, we propose a novel deep learning model, TCRaman, which integrates transfer learning and contrastive learning for Raman spectra data of urine, to detect urinary glucose, protein and predict kidney disorders. TCRaman first utilizes a pretrained contrastive learning model on a large labeled Raman spectral dataset of bacteria [34], then the pretrained model is finetuned on clinical urine Raman spectral data. Contrastive learning discriminates representations by maximizing the similarity between similar samples and minimizing the similarity between dissimilar samples [47–49], and contrastive learning with focal loss can better tackle the imbalanced data [50,51]. In summary, our main contributions of this paper are as follows.

- TCRaman integrates transfer learning and contrastive learning for detection of glucose, protein, and prediction of kidney disorders using Raman spectral data of urine, aiming to address the challenges posed by high-dimensionality, imbalance and sample-scarcity of biochemical Raman spectral data.
- TCRaman is a transfer learning framework for Raman spectral data, which can be applied to multiple downstream tasks using single pretrained model, significantly extending the range of applications of deep learning to the diverse tasks of Raman spectral data.
- Contrastive learning with focal loss helps the model to capture discriminative, meaningful low-dimensional representations from the imbalanced urine Raman spectra data, enhancing the performance on multiple downstream tasks.

The experiments demonstrate that TCRaman achieves 91% accuracy in detecting glucose and protein, as well as 95% accuracy in predicting kidney disorders, emphasizing its superiority compared with current models, providing a promising way for accurate, rapid, and cost-effective detection for spectral data of biochemical samples.

Table 1
Details of urine Raman spectral dataset.

Dataset	Training		Testing	
Label	N	P	N	P
Glucose	1080	228	681	246
Protein	1083	225	702	225
Kidney disorders	1038	270	807	120

2. Materials and methods

2.1. Raman spectral data acquisition

The urine samples are provided by the First People's Hospital of Foshan. 2235 urine samples are collected from September, 2023 to March, 2024. As far as we know, this is the largest dataset for Raman spectral data analysis of urine. The 1308 samples collected in 2023 are adopted as training set, and the rest 927 samples collected in 2024 are adopted as testing set. The qualitative labels of urine glucose and protein are accessed by urine analyzer (UC-3500, Sysmex, Japan). Urine glucose concentrations higher than 2.8mmol/L are diagnosed as abnormal, and urine protein concentrations higher than 0.15g/L are diagnosed as abnormal. Details of the dataset are listed in Table 1. Label N denotes negative (normal) samples, and label P denotes positive (abnormal) samples. It is obvious that the dataset is imbalanced, where negative samples are much more than positive samples.

The Raman spectral data of samples are acquired following our previous studies [23,24]. The entire Raman spectroscopy system is placed in a temperature-controlled cabinet (SNKM-250L, SNUGEN, China) which maintains the constant temperature at 25 ±0.1°C. The Raman spectral acquisition is performed using a NOVA2S Raman spectrometer (7P-785-R, Ideaoptics, China). equipped with a 785nm-fiber laser source (fL-785-T05-R, Ideaoptics, China) operating at the power maxima of 523 mW. The Raman shift ranging from 400 to 2000 cm^{-1} is recorded at a resolution of 6 cm^{-1}. Each Raman spectrum is preprocessed by background correction using adaptive iterative reweighted penalized least squares (airPLS) [52] and 0–1 normalization. The dimension of each Raman spectrum is 360.

2.2. Transfer learning

As a transfer learning model, the pretraining procedure and finetuning procedure are certainly implemented in TCRaman. As shown in Fig. 1(a), TCRaman first utilizes a pretrained contrastive learning model for the classification task of a large labeled Raman spectral dataset of bacteria [34]. The contrastive learning model includes two modules: representation module and classifier module. The representation module takes the bacteria Raman spectral data X as input, to learn the bacteria Raman spectral representation Z. The classifier module adopts representation Z for bacteria classification, and outputs the predicted bacteria categories Y.

The bacterial Raman spectral dataset in pretraining procedure includes 30 categories of bacteria, totally 60000 samples for training, and 3000 samples for testing. The range of Raman shift is 382–1792 cm^{-1}. The dimension of each Raman spectrum is 1000.

Both bacteria and urine Raman spectra datasets share common underlying features and patterns of biological liquid samples, despite their different sources and contexts. Both bacteria and urine involve biological samples and the analysis of microbial communities, allowing the model trained on the bacterial Raman spectra dataset to capture relevant features that are transferable to the urine Raman spectra dataset. This connection ensures that the model can migrate effectively, retaining its predictive power and generalizability while adapting to the specific characteristics of the new dataset.

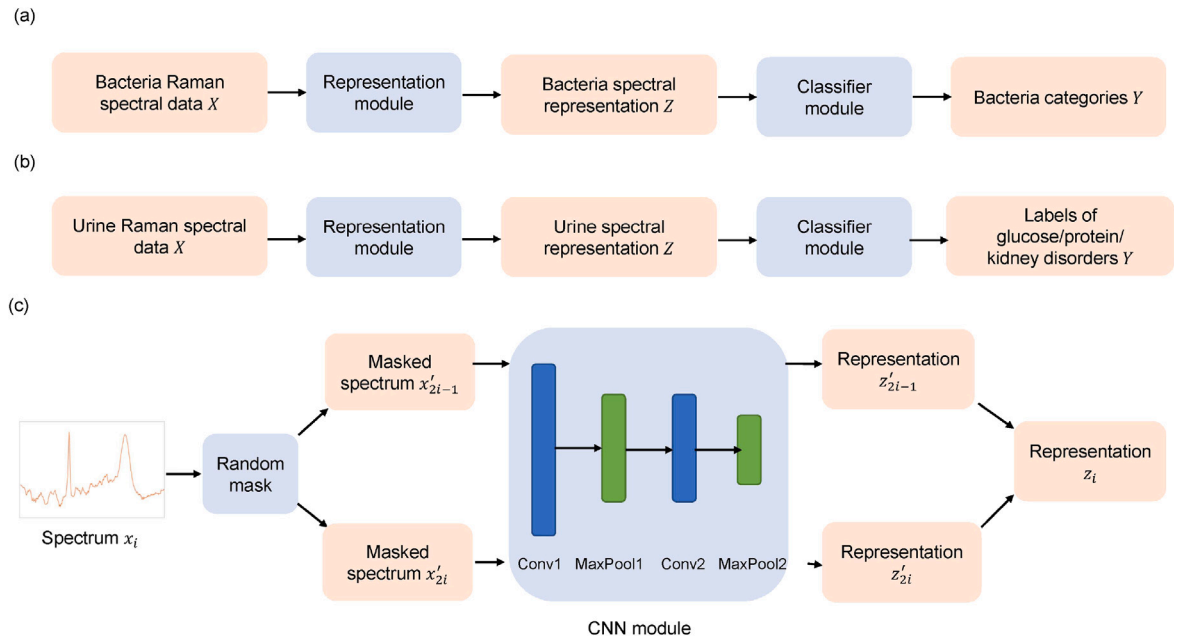


Fig. 1. The schematic of TCRaman. (a) Pretraining procedure. (b) Finetuning procedure. (c) Representation module of contrastive learning model.

In order to ensure the consistency of bacterial spectral dimension and urine spectral dimension, the bacterial spectral dimension is adjusted to 360 by linear interpolation. Then, as shown in Fig. 1(b), the pretrained model is finetuned on our collected urine Raman spectral data X. Similar to pretraining procedure, the representation module aims to learn the urine Raman spectral representation Z. The classifier module adopts representation Z to output the predicted label Y for the detection of both glucose and protein, and the prediction of kidney disorders.

2.3. Contrastive learning

2.3.1. Representation module

Let x_i denotes the ith spectrum of Raman spectral data X. After data augmentation, x_i can generate two different samples, x'_{2i-1} and x'_{2i}. As shown in Fig. 1(c), the representation z'_{2i-1} and z'_{2i} can be learned from x'_{2i-1} and x'_{2i}, respectively. Contrastive representation learning aims to maximizing the similarity between z'_{2i-1} and z'_{2i}, while minimizing the similarity between z'_{2i-1} (or z'_{2i}) and all other samples, to obtain the final representation z_i.

Random mask is a powerful way for spectral data augmentation. Following the spectral masking method proposed in [32], let m denotes the dimension of spectrum (in this paper $m = 360$), a is an integer sampled randomly from a uniform distribution of $[0, m)$, and d is a number sampled randomly from a uniform distribution of [0.3,0.7]. Let

$$b = \begin{cases} m & a > (1-d)m \\ \mathrm{int}(a + dm) & a \le (1-d)m \end{cases} \tag{1}$$

where int denotes integer rounding. Masking the $[a, b)$ interval of this spectrum is performed by setting the intensity values in the interval to 0. After performing double different random masks on x_i, we can obtain two different masked spectra x'_{2i-1} and x'_{2i}. An example illustrating spectrum x_i before masking, and spectra x'_{2i-1} and x'_{2i} after masking, is shown in Fig. S1.

Then, through the convolutional neural networks (CNN) module, the representation z'_{2i-1} and z'_{2i} can be learned from x'_{2i-1} and x'_{2i}, respectively. The CNN module consists of convolutional layer Conv1 and Conv2, and maximum pooling layer MaxPool1 and MaxPool2. The convolutional layer can be regarded as a filter to learn salient features in high-dimensional spectral data while suppressing noise. The maximum pooling layer can reduce the dimension of the data by taking the maximum value in the pooling window region, so as to learn the low-dimensional meaningful representations. Following previous research that adopts CNN to Raman spectra classification [35], the pooling size is set to 2. After one pooling layer, the feature dimension is reduced to 1/2 of the input data. Hence, the dimension of z'_{2i-1} and z'_{2i} is $360/2^2 = 90$.

Let n denotes the number of samples in original Raman spectral data X, for all $p = 1, 2, \ldots, 2n$ and $q = 1, 2, \ldots, 2n$, the contrastive loss [48] can be defined as

$$\ell(p, q) = -\log \frac{\exp\left(s(z'_p, z'_q)/\tau\right)}{\sum_{r=1}^{2n} \beta_{pr} \exp\left(s(z'_p, z'_r)/\tau\right)} \tag{2}$$

where s denotes cosine similarity function

$$s(z'_p, z'_q) = \frac{z'^T_p z'_q}{\|z'_p\| \cdot \|z'_q\|} \tag{3}$$

β denotes inequality indicator function

$$\beta_{pr} = \begin{cases} 1 & p \neq r \\ 0 & p = r \end{cases} \tag{4}$$

and τ is a hyperparameter. The contrastive loss of the whole dataset is

$$L_c = \frac{1}{2n} \sum_{i=1}^{n} (\ell(2i-1, 2i) + \ell(2i, 2i-1)) \tag{5}$$

Note that $\ell(2i-1, 2i) \neq \ell(2i, 2i-1)$. The explanation of contrastive loss is in Supporting Information. Minimizing contrastive loss L_c aims to maximizing the similarity between z'_{2i-1} and z'_{2i}, while minimizing the similarity between z'_{2i-1} (or z'_{2i}) and all other samples, to obtain the final representation

$$z_i = \frac{z'_{2i-1} + z'_{2i}}{2} \tag{6}$$

2.3.2. Classifier module

Classifier module is a two-layer multi-layer perceptron (MLP) that adopts representation Z as input to output the predicted label Y. The input dimension of classifier module is 90. The hidden dimension is 32. The output dimension is the number of classes. In pretraining

procedure, the output dimension is 30 for discriminating 30 categories of bacteria. In finetuning procedure, the last layer of classifier module is replaced by a new layer, to ensure the output dimension is 2 for binary discrimination of glucose, protein or kidney disorders. The discriminative loss for classification is focal loss for tackling imbalanced classification [53].

$$L_f = -\frac{1}{n}\sum_{i=1}^{n}(1-p_i)^{\gamma}\log p_i \tag{7}$$

Here, p_i denotes the predicted probability of the ith spectrum, computed by $p_i = \hat{y}_i^T y_i$, where $\hat{y}_i$ and y_i denote the true label and predicted label of the ith spectrum, respectively, γ is hyperparameter. The explanation of focal loss is in Supporting Information. Finally, the total loss

$$L = L_c + \alpha L_f \tag{8}$$

where α is a hyperparameter to balance focal loss and contrastive loss. Herein, L_c is an unsupervised contrastive learning (UCL) loss, and the integration of discriminative loss and contrastive loss leads to supervised contrastive learning (SCL). Recent studies show that SCL can enhance the representation learning ability of UCL [54,55]. Instead of commonly used cross entropy loss as discriminative loss, SCL with focal loss can better tackle the imbalanced classification [50,51]. The explanation is in Supporting Information.

2.4. Model training

We adopt PyTorch (https://pytorch.org/) to implement the model, and apply an Adam optimizer with learning rate of 0.01 and weight decay of 10^{-5} to train the model. The model is trained on NVIDIA GeForce RTX 3060 GPU with 12 GB memory. The pretraining procedure and finetuning procedure are trained by 100 iteration times, respectively. The activation function of the last layer of classifier module is set as softmax function, while the activation functions in convolutional layers are set as rectified linear unit (ReLU) function. The details of layers in TCRaman are shown in Table S1.

3. Results and discussions

3.1. Hyperparameters tuning

There are seven key hyperparameters in TCRaman model: hyperparameter τ in contrastive loss Eq. (2), hyperparameter γ in focal loss Eq. (7), hyperparameter α in total loss Eq. (8), numbers of convolutional layers in representation module, size of convolutional kernels in representation module, hidden dimension of the classifier module, and the learning rate of Adam optimizer. Hyperparameters tuning are conducted on urine Raman spectra dataset during finetuning. As random search of hyperparameters can achieve best performance with time-efficiency [56], we adopt random search in a set of hyperparameters (see Table 2). Random search is performed by adopting 5-fold cross validation on the training set using the accuracy of kidney disorders prediction as validation metric. For each fold, random search is performed for 30 times, and the optimal hyperparameter values among these 30 randomizations are chosen. After that, the best hyperparameters are set as $\tau = 0.2, \alpha = 0.2, \gamma = 2$, the numbers of convolutional layers in representation module is 2, the size of convolutional kernels in representation module is 5, the hidden dimension of the classifier module is 32, and the learning rate is 0.01. The best hyperparameters after hyperparameters tuning are adopted on testing set for further evaluation of the model.

In order to verify the performance of the best hyperparameters obtained by random search, we changed the value of one of the best hyperparameters each time and fixed the other hyperparameters to verify the accuracy of TCRaman's prediction of glucose, protein and kidney disorders, as shown in Fig. S2–S8. The results show that changing the value of any of the hyperparameters reduces the accuracy of all three prediction tasks (predicting glucose, protein and kidney disorders). These results show that the hyperparameter values obtained by random search are the best candidates.

Table 2
Candidate hyperparameters.

Name	Candidate values	Best value
τ	0.1,0.2,0.4,0.8	0.2
α	0.1,0.2,0.4,0.8	0.2
γ	1,2,4	2
Number of Conv layers	1,2,3	2
Kernel size	3,5,7	5
Hidden dimension	16,32,64	32
Learning rate	0.001,0.01,0.1	0.01

Table 3
Comparison on glucose detection.

Model	Acc	Spe	Pre	Rec	F1	AUC
PCA-LDA	0.762	0.779	0.539	0.715	0.615	0.804
PLS-DA	0.735	0.740	0.501	0.723	0.592	0.798
SVM	0.739	0.764	0.458	0.657	0.540	0.771
RF	0.860	0.885	0.474	0.691	0.562	0.863
XGBoost	0.803	0.866	0.628	0.626	0.627	0.838
LightGBM	0.837	0.856	0.423	0.708	0.529	0.861
CNN	0.818	0.810	0.406	0.875	0.556	0.903
LSTM	0.802	0.823	0.570	0.733	0.642	0.857
Transformer	0.870	0.916	0.702	0.850	0.775	0.910
TCRaman	0.912	0.915	0.793	0.902	0.844	0.967

3.2. Comparison with other methods

Accuracy (Acc), specificity (Spe), precision (Pre), recall (Rec), F1 score, and the area under receiver operating characteristic curve, known as AUC, are commonly used metrics for binary classification. The higher values of all the above six metrics indicate superior performance. In addition, the confusion matrix (Fig. 2) is also used to visualize the results of classification.

We compared our proposed method, TCRaman, with commonly used (1) machine learning models: principal component analysis-linear discriminant analysis (PCA-LDA), partial least squares discriminant analysis (PLS-DA), support vector machine (SVM), random forest (RF), XGBoost and LightGBM, (2) deep learning models: convolutional neural networks (CNN) [30], long short-term memory (LSTM) [35], and Transformer [33]. The Transformer is implemented following [33]. The hyperparameters of the other compared machine learning and deep learning models are tuned by random search similar to hyperparameters tuning of TCRaman to achieve better performance, and the details are listed in Table S2. The structural parameters of the compared CNN is consistent to the CNN module in the representation module of TCRaman, where the input dimension and output dimension of the CNN are 360 and 90, respectively. Then, the output of the compared CNN is adopted as the input of the classifier module consistent to TCRaman. We also evaluated CNN, LSTM, Transformer and TCRaman at different epochs, as shown in Fig. S9. The results show that all three models perform best at epoch 100. Too many epochs may cause overfitting on the dataset.

The results of metrics on the detection of glucose, protein and kidney disorders are listed in Table 3, Table 4 and Table 5, respectively. The comparison results demonstrate that TCRaman can achieve best performance in all three tasks: detection of glucose, protein, and prediction of kidney disorders. TCRaman achieves 0.912 accuracy on the detection of glucose, 0.908 accuracy on the detection of protein, and 0.954 accuracy on the prediction of kidney disorders, highlighting the effectiveness of our proposed method in detecting urine Raman spectra. As TCRaman preforms best at all six metrics (Acc, Spe, Pre, Rec, F1 and AUC), the results and the confusion matrices (Fig. 2) suggest that TCRaman is capable of tackling imbalanced data. Besides, Transformer can outperform all other models except TCRaman.

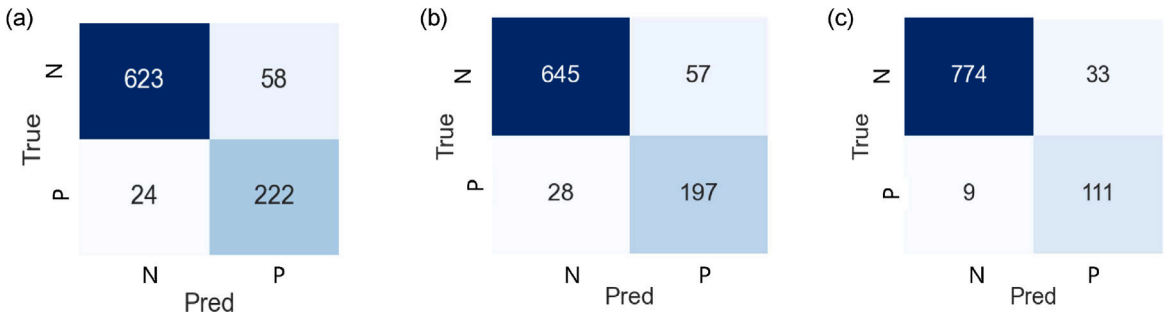


Fig. 2. The confusion matrix of TCRaman results on the detection of (a) glucose, (b) protein, and (c) kidney disorders.

Table 4
Comparison on protein detection.

Model	Acc	Spe	Pre	Rec	F1	AUC
PCA-LDA	0.786	0.796	0.543	0.755	0.631	0.853
PLS-DA	0.795	0.820	0.560	0.715	0.628	0.844
SVM	0.742	0.728	0.540	0.778	0.638	0.817
RF	0.813	0.785	0.600	0.890	0.716	0.910
XGBoost	0.801	0.777	0.613	0.855	0.714	0.894
LightGBM	0.848	0.875	0.648	0.759	0.699	0.889
CNN	0.873	0.878	0.681	0.861	0.761	0.934
LSTM	0.856	0.830	0.639	0.937	0.760	0.959
Transformer	0.882	0.912	0.698	0.845	0.791	0.936
TCRaman	0.908	0.919	0.776	0.876	0.823	0.966

Table 5
Comparison on kidney disorders prediction.

Model	Acc	Spe	Pre	Rec	F1	AUC
PCA-LDA	0.811	0.805	0.393	0.850	0.538	0.864
PLS-DA	0.813	0.814	0.392	0.808	0.528	0.854
SVM	0.746	0.745	0.484	0.747	0.587	0.820
RF	0.879	0.881	0.519	0.866	0.650	0.925
XGBoost	0.845	0.867	0.439	0.700	0.540	0.865
LightGBM	0.848	0.874	0.648	0.759	0.699	0.889
CNN	0.882	0.871	0.743	0.911	0.819	0.950
LSTM	0.873	0.872	0.675	0.875	0.762	0.939
Transformer	0.895	0.906	0.788	0.891	0.796	0.954
TCRaman	0.954	0.959	0.771	0.925	0.841	0.982

3.3. Ablation studies

To demonstrate the effectiveness of transfer learning and contrastive learning, respectively, we also conduct ablation studies for TCRaman. We design two models: TRaman and CRaman.

- TRaman is only with transfer learning, i.e. the total loss is only focal loss.
- CRaman is only with contrastive learning, i.e. the model is trained on urine spectra data directly, without pretraining procedure.

The results are shown in Table 6. Notably, either TRaman or CRaman is also superior to current models, highlighting the necessity of adding transfer learning and contrastive learning to current deep learning models, suggesting that the integration of transfer learning and contrastive learning can improve the ability of representation learning. To further demonstrate the validity of transfer learning, ablation studies are conducted and are shown in Table S3. Moreover, the results of ablation studies on contrastive learning, sampling and pooling in CNN, are shown in Fig. S10–S12, respectively. The details for ablation studies on TCRaman with or without sampling are provided in Supporting Information.

3.4. Interpretability analysis

Some key peaks of the Raman spectrum are associated with specific chemical structures. Gradient-weighted class activation mapping (Grad-CAM) [57] is a common method for calculating the importance of different dimensions in neural networks. Following previous studies [31,32], to interpret which peaks of the spectrum are more important in TCRaman training, we adopt Grad-CAM to calculate the importance of TCRaman on peaks of different spectra, as shown in Fig. 3. Compared to classical analytical methods identifying several key peaks in the Raman spectrum, TCRaman is more inclined to make full use of all spectral information.

However, it is still important to identify specific Raman peaks with chemical significance [58]. Raman peak at 937 cm^{-1} is correspond to C-O-C stretching of carbohydrates (especially glucose) [59]. Raman peak at 1558 cm^{-1} is correspond to ν(CN) and δ(NH) of amide II, which is recognized as the signature of protein [60,61]. As shown in Fig. 3(a) and (b), the highest values of Grad-CAM on the detection of glucose or protein appear at the signature peak of these two biomolecules, respectively. Statistical analysis of these two peaks are shown in Fig. S13–S15.

Notably, in Fig. 3(c), Grad-CAM achieves high values at both signature peaks of glucose and protein on the prediction of kidney disorders. This phenomenon indicates that more comprehensive information has been captured on the prediction of kidney disorders, including information about glucose peaks and protein peaks, as well as more characteristic patterns hidden in the Raman spectral data. Hence, from Table 3 to Table 5, the prediction accuracy of kidney disease is higher than that of glucose or protein. As most existing methods of disease diagnosis based on urine Raman spectroscopy (see Introduction) are proposed by comparing the characteristic peaks of key biomolecules in urine between negative (normal) and positive (patient) samples, direct prediction of diseases with an end-to-end, data-driven approach can achieve better results through more comprehensive learning.

Furthermore, the mean spectra of negative and positive samples are shown in Fig. 4(a)–(c). More spectra of negative and positive samples are shown in Fig. S16 (a)–(c). It is difficult to observe the slight differences between spectra of negative and positive samples. Fig. 4(d)–(f) shows the t-distributed stochastic neighbor embedding (T-SNE) visualization results of representation Z learning on the detection of (d) glucose, (e) protein, and (f) kidney disorders. Most of the positive and negative sample points are in different locations, illustrating the discriminative ability of representations on all three tasks. Notably, on the representations learned in the kidney disorders prediction task, points of the same class (negative or positive) are more concentrated, while points of different classes are further apart. This phenomenon further confirms that the model learns more information in predicting kidney disease, so these representations are more discriminative.

3.5. Case studies

Case studies are conducted by analyzing the results on joint detection of glucose and protein. Usually, patients who tested positive for both urinary glucose and protein are with a higher risk of kidney disorders. As shown in Table S4, either glucose or protein positive is not significantly correspond to kidney disorders positive ($p>0.05$), while

Table 6
Ablation studies.

Detection	Model	Acc	Spe	Pre	Rec	F1	AUC
Glucose	TRaman	0.885	0.891	0.708	0.866	0.779	0.941
	CRaman	0.890	0.896	0.718	0.870	0.787	0.942
	TCRaman	0.912	0.915	0.793	0.902	0.844	0.967
Protein	TRaman	0.891	0.898	0.722	0.866	0.787	0.942
	CRaman	0.886	0.892	0.708	0.865	0.779	0.940
	TCRaman	0.908	0.919	0.776	0.876	0.823	0.966
Kidney disorders	TRaman	0.893	0.873	0.707	0.955	0.813	0.970
	CRaman	0.905	0.916	0.768	0.871	0.816	0.964
	TCRaman	0.954	0.959	0.771	0.925	0.841	0.982

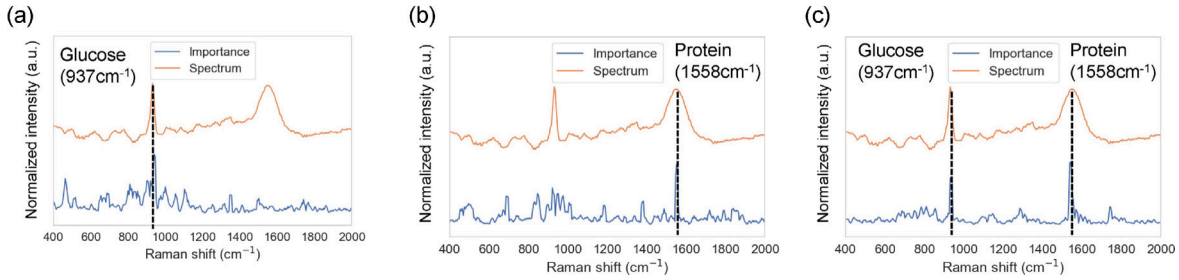


Fig. 3. Grad-CAM results of (a) glucose, (b) protein and (c) kidney disorders.

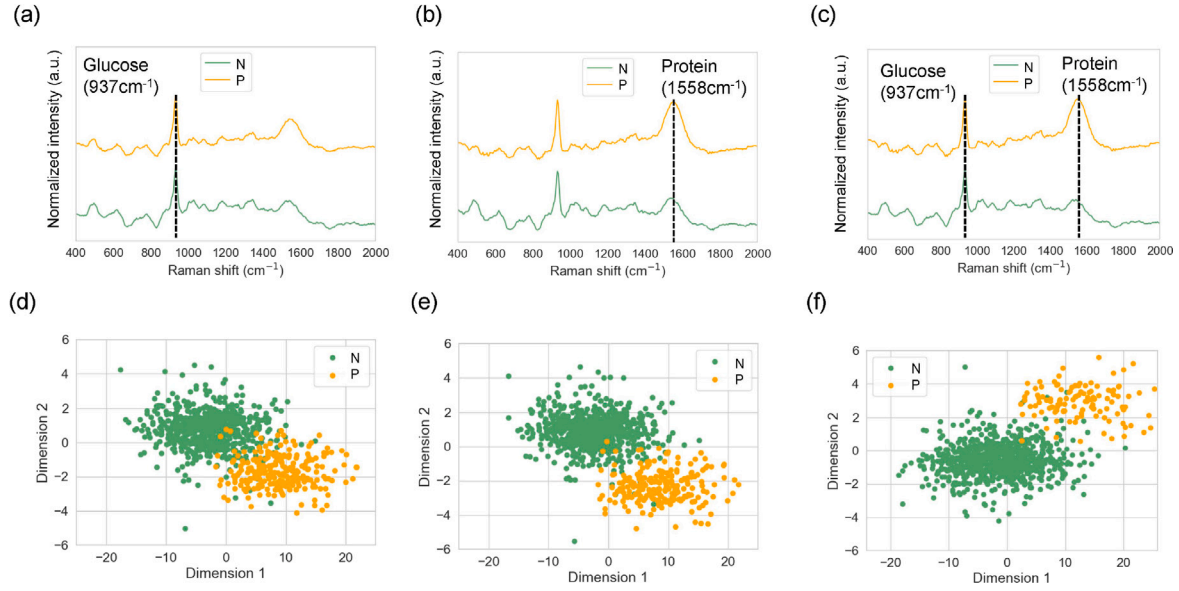


Fig. 4. (a–c) The mean spectra of negative and positive samples of (a) glucose, (b) protein, and (c) kidney disorders. (d–f) The T-SNE visualization results of representation Z learning on the detection of (d) glucose, (e) protein, and (f) kidney disorders.

double-positive is significantly correspond to kidney disorders positive ($p=0.002$).

To evaluate TCRaman's ability of identifying double-positive samples, case studies are conducted on joint detection of glucose and protein. The confusion matrix combining TCRaman's predictions for urinary glucose and protein are shown in Fig. 5(a). Label N/N denotes double-negative, label N/P denotes that glucose is negative and protein is positive, label P/N denotes that glucose is positive and protein is negative, label P/P denotes double-positive. TCRaman achieves 0.831 accuracy on this multi-classification task. Notably, for the 69 double-positive samples, the model correctly predicted 56 of them to be double-positive, while without misdiagnosing any of them as double-negative. As shown in Fig. 5(b), the results illustrate TCRaman's ability to identify those few but special samples.

Besides, we have also conducted case studies by computing TCRaman's accuracy performance of different age groups. The results are shown in Table S5, demonstrating TCRaman's robustness on different age groups.

4. Conclusions

Raman spectroscopy exhibits significant promise in diagnosing kidney disorders by detecting vital biomolecules in urine samples, especially glucose and protein. However, the high-dimensionality, imbalance, and scarcity of data still remain challenges for machine learning-assisted urinary Raman spectrum detection. To tackle these challenges, we propose TCRaman, a novel deep learning model that integrates transfer learning and contrastive learning for urine detection utilizing Raman spectral data. TCRaman employs a pretrained contrastive learning model on a vast labeled Raman spectral dataset of bacteria, enhancing its capacity to extract meaningful low-dimensional representations from raw high-dimensional Raman spectral data. Subsequently,

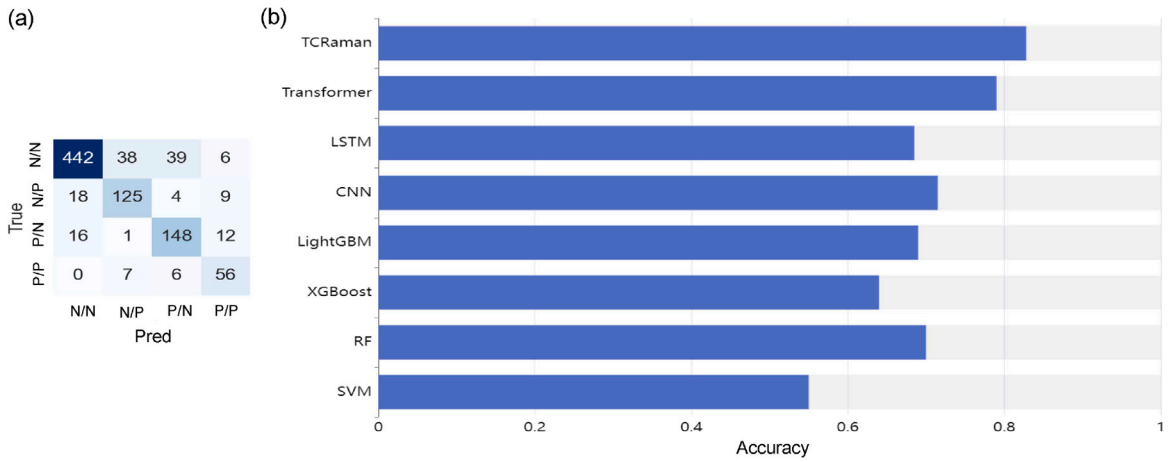


Fig. 5. (a) The confusion matrix of TCRaman and (b) the comparison results on joint detection of glucose and protein.

this pretrained model is finetuned on clinical urine Raman spectral data, overcoming the limitation of data imbalance and scarcity through the transfer learning framework. Experimental results demonstrate the superiority of TCRaman over existing state-of-the-art models. TCRaman achieves 91% accuracy in detecting both glucose and protein, and 95% accuracy in predicting kidney disorders, highlighting the effectiveness of our proposed method in analyzing urine Raman spectra. The Grad-CAM and T-SNE visualization results validates the interpretability of TCRaman, while case studies on glucose–protein joint detection illustrate TCRaman's ability to identify those few but special double-positive samples.

However, there are also some limitations in TCRaman. Firstly, the resolution of the existing instrument is relatively low when used to detect biochemical samples such as urine, our future work will focus on optimizing and improving the resolution of the instrument and developing corresponding machine learning models to detect more features such as key peaks from the spectrum of biochemical samples. Secondly, in contrastive learning of TCRaman, we adopt basic cosine similarity to measure the similarities among samples. Recently, there are some deep learning models adopting other similarity metrics for a better measurement, such as adjusted cosine similarity [62] and Pearson correlation coefficient (PCC) [63]. We will also adopt other similarity metrics to improve our contrastive learning model in our future work. Thirdly, in our study, the classification of kidney diseases as normal or abnormal was based on the compiled medical records of the patients. In our future works, we plan to conduct more in-depth data mining of these records, such as different kidney disease severities, to refine the classification model of kidney diseases.

In conclusion, TCRaman proves that the integration of transfer learning and contrastive learning can provide a promising avenue for accurate, rapid, and cost-effective detection of diverse tasks in biochemical spectral data. Therefore, TCRaman is potential to be applied for other spectroscopy techniques such as infrared spectroscopy or fluorescence spectroscopy, and other biochemical samples, such as blood, bacteria or foods.

CRediT authorship contribution statement

Zhuangwei Shi: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiale Wang:** Validation, Resources, Investigation, Formal analysis, Data curation. **Yunhao Su:** Visualization, Investigation, Formal analysis. **Xiaohong Liang:** Software, Investigation, Formal analysis. **Jianchen Zi:** Writing – review & editing, Visualization, Supervision, Project administration, Investigation. **Chenhui Wang:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Hai Bi:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Xia Xiang:** Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.chemolab.2025.105384.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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